Knowledge-Guided Hierarchical CVE-to-CWE Mapping using HMCN and Cross-Attention*

Yeong-jin Kim, Taewoong Kwon and Jun Lee
Korea Institute of Science and Technology Information(KISTI), Daejeon, Republic of Korea
{kyj, taewoong.kwon, jun.lee}@kisti.re.kr

Abstract

CVE-to-CWE mapping is important for vulnerability root-cause analysis and automated security response. However, most existing approaches rely on flat multi-label classification structures that do not sufficiently reflect hierarchical relationships among CWE categories. In this paper, we propose a knowledge-guided CVE-to-CWE mapping framework based on Hierarchical Multi-label Classification Networks (HMCN) and cross-attention mechanisms. The proposed framework reflects the VIEW-1000 CWE hierarchy and incorporates semantic information through a static CWE knowledge base. The proposed model progressively propagates hierarchical contextual information across multiple HMCN levels while repeatedly applying cross-attention using CWE semantic embeddings. In addition, global classification loss, local hierarchical loss, and hierarchy-aware constraint loss are jointly optimized to improve hierarchical consistency. Preliminary experiments using NVD CVE records and MITRE CWE data show that the proposed approach reduces hierarchy violations while maintaining competitive classification performance.

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1 Introduction

CVE (Common Vulnerabilities and Exposures) and CWE (Common Weakness Enumeration) are widely used standards for vulnerability analysis. Existing automated CWE mapping approaches mainly formulate the task as a flat text classification problem using CVE descriptions. Unlike existing flat classification approaches and general-purpose hierarchical classification frameworks, the proposed method incorporates CWE semantic knowledge through cross-attention mechanisms to improve hierarchical consistency. However, such methods often fail to reflect hierarchical relationships among CWE categories, resulting in hierarchy violations where child categories are predicted without corresponding parent categories. To construct the hierarchical classification structure, the VIEW-1000 CWE hierarchy was parsed from MITRE CWE XML data by extracting parent-child relationships and organizing CWE categories into multiple hierarchy levels.

2 Proposed Method

The proposed framework adopts an HMCN-based hierarchical architecture reflecting the VIEW-1000 CWE hierarchy. Hierarchical blocks progressively propagate contextual information across multiple levels. A static CWE knowledge base is constructed using CWE descriptions provided by MITRE. Cross-attention is repeatedly applied at each hierarchical level using CWE semantic embeddings to reinforce semantic context and improve semantic alignment between vulnerabilities and CWE categories. The framework jointly optimizes three objectives:

- Global Loss: overall classification objective;
- Local Loss: hierarchy-level classification objective.
- Hierarchy-aware Loss: penalty for hierarchy violations

3 Experimental Setup

Experiments were conducted using CVE records collected from NVD and CWE data provided by MITRE. The dataset consisted of 95,239 CVEs and 682 CWE categories. A time-based split was applied using CVEs from 2016–2024 for training and CVEs from 2025 for evaluation.

4 Results

Experiments show improved hierarchical consistency with competitive performance. The hierarchy-aware optimization reduced hierarchy violations from 483 to 22 while maintaining competitive classification performance, although Recall tended to decrease as hierarchy constraints increased, indicating a trade-off between hierarchical consistency and classification performance. In addition, cross-attention with CWE knowledge improved semantic alignment.

5 Conclusion and Future Work

This paper presented a knowledge-guided hierarchical framework for automated CVE-to-CWE mapping using HMCN and cross-attention mechanisms. Future work includes traversal-based hierarchical mapping and integrating large language models (LLMs) for enhanced semantic reasoning.

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